



Examining Age-Related Differences in Neural Responses to Trust



Abraham Dachs¹, Cooper J. Sharp¹, Enes Yanilmaz¹, James B. Wyngaarden III¹, Ori Zaff², Tia Tropea¹, Daniel Sazhin¹, Rita Ludwig², Tania Giovannetti¹, Dominic S. Fareri³, David V. Smith¹

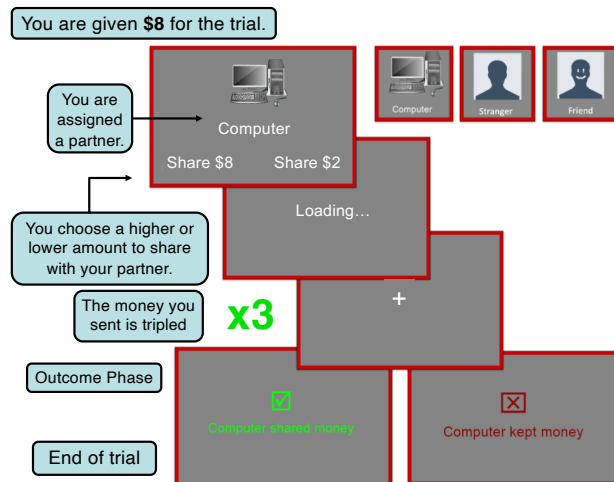
1. Temple University, Philadelphia, PA; 2. University of Pennsylvania, Philadelphia, PA; 3. Adelphi University, Garden City, NY

1. Introduction

- Social relationships influence our behaviors and may modulate our neural responses to reward. (Fareri et al., 2015)
- The effect of relationships on behavior and reward activation may change as we age. (Fareri et al., 2022)
- It remains unclear whether age-related differences in responses to trust are associated with maladaptive outcomes such as financial exploitation.

We investigated whether social trust decisions change as we age, and if those changes could be related to financial exploitation.

2. Methods



N = 62 Completed the Inclusion of Other in Self Scale (IOS) and the Older Adults Financial Exploitation Measure (OAFEM).

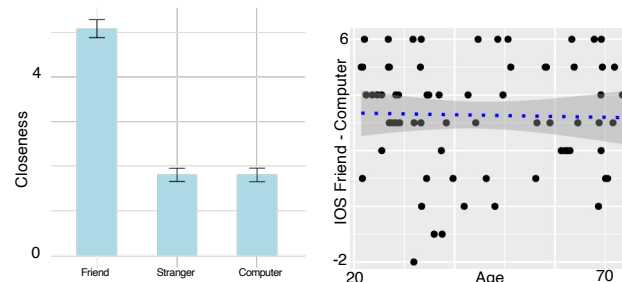
Whole brain results were thresholded at $Z > 3.1$ and Family Wise Error corrected at $p > .05$



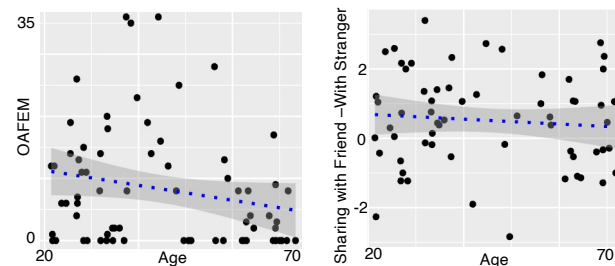
Scan the QR code to see the Pre-registration!

This study was supported by NIH grant RF1-AG67011 (David V. Smith). Contact information: Avi.Dachs@temple.edu, Cooper.Sharp@temple.edu

3. Age Related Behavioral Effects

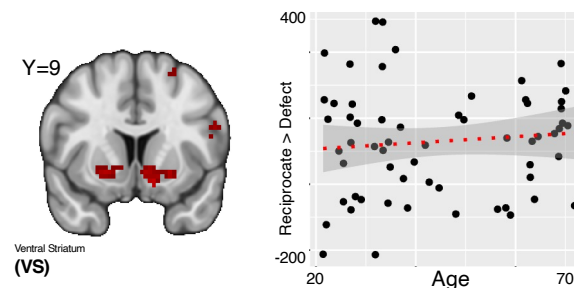


Participants are closer with their friend than with the stranger and computer partners, however, there is no relationship between age and closeness ratings.



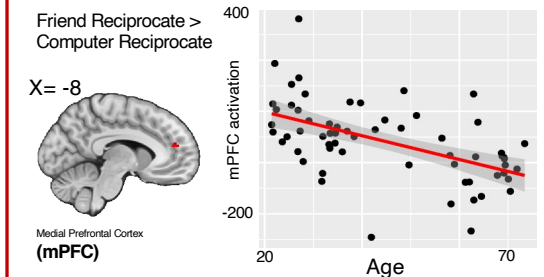
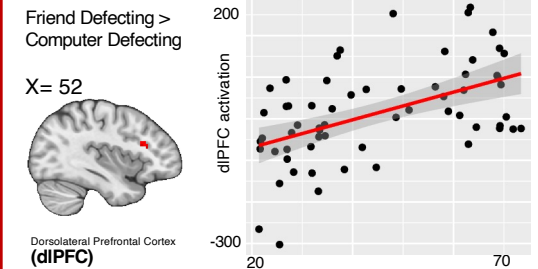
We found no age-related effects across financial exploitation measures or investment behavior

4. Age-Related Differences in Striatal Response?



We found activation in the VS when participants were presented with a reciprocate outcome as opposed to a defect outcome, however, there was no significant interaction with age.

5. Age Related Differences in Partner Response



We found that friend defecting resulted in increased dlPFC activation in older adults compared to younger adults.

We also found that friend reciprocating resulted in reduced mPFC activation in older adults compared to younger adults.

6. Conclusions and Future Directions

- Our preliminary results indicate that **reciprocation elicited activation in the ventral striatum**.
- Additionally, we find that the **dlPFC and mPFC activation differs between older and younger adults** relative to social context.
- Future analyses will further examine age-related differences in neural responses to monetary outcomes relative to social context and risk of financial exploitation.